REVENUE LEAKAGE INTELLIGENCE

Executive analytical report

A quantified assessment of contracted-to-collected revenue erosion, customer risk, commercial discipline, and intervention priorities.

18.2%	$6.0 million	$4.2 million	48
of contracted revenue not collected	observed 24-month revenue leakage	scored ARR exposure	high and critical accounts

Analysis period: January 2024 to December 2025
Customer base: 500 accounts
Annual recurring revenue: $19.2 million

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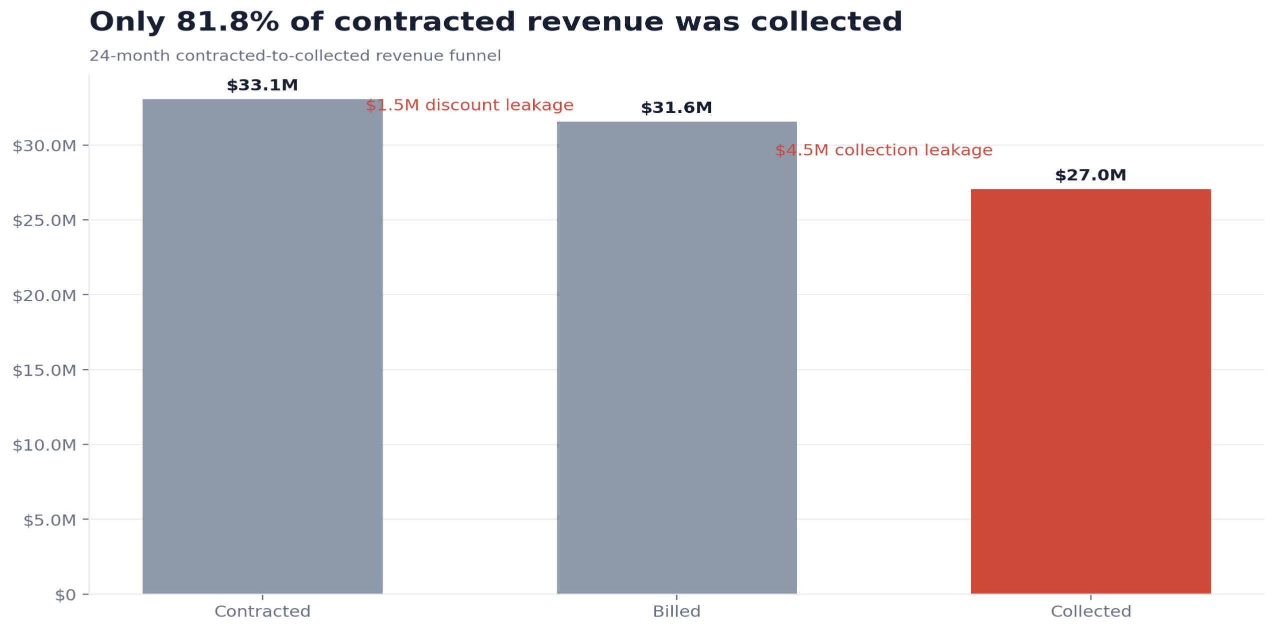
The report is structured around the decisions management needs to make: quantify leakage, identify its causes, locate concentrated exposure, and define an operating response.

Executive summary

Across the 24-month observation period, the business contracted $33.1 million of revenue and collected $27.0 million. The difference is $6.0 million, equivalent to 18.2% of contracted revenue. Discounting accounts for $1.5 million; collection failures account for $4.5 million. The scale and persistence of the gap indicate an operating control problem rather than isolated account exceptions.

The scored risk model identifies $4.2 million, or 21.7% of ARR, as exposed. There are 1 critical and 47 high-risk accounts. The top 20 accounts represent 24.7% of all scored revenue at risk, which makes a focused intervention programme practical.

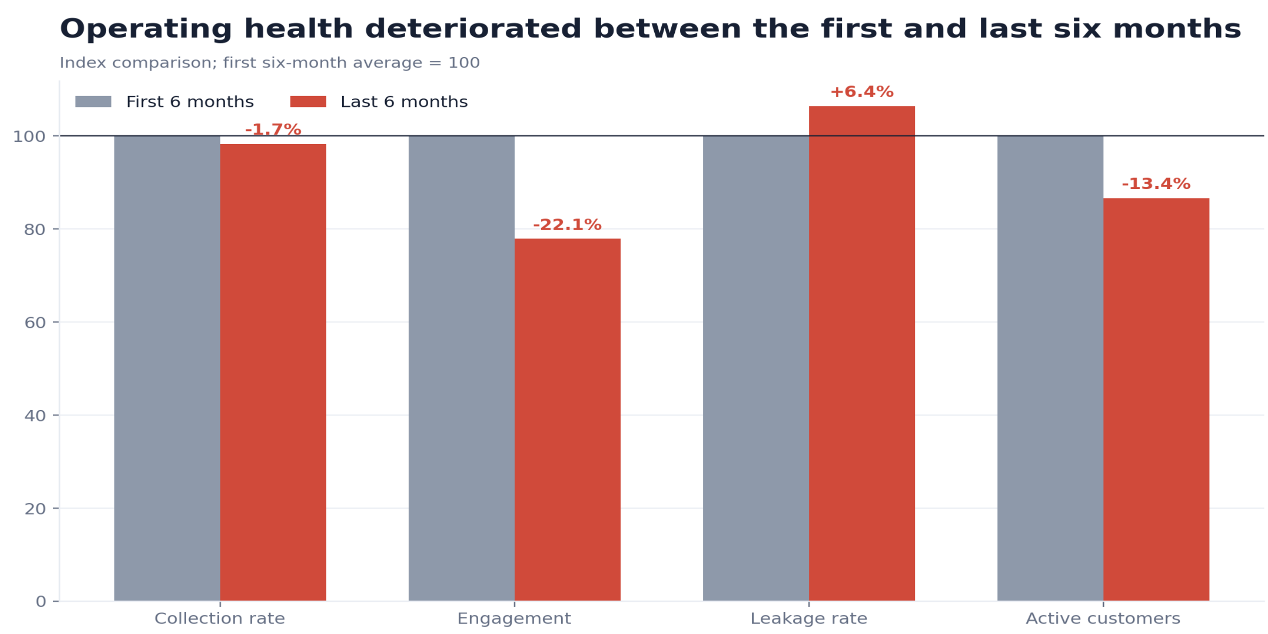


Management should treat the collection gap as the first control priority. Payment failures contribute the largest share of weighted composite risk and the collection rate weakened between the first and last six months. Discount controls are the second immediate opportunity because the loss is measurable, concentrated by sales representative, and preventable before invoices are issued.

Executive summary: decision priorities

Priority	Evidence	Required decision
1. Recover collections	$4.5 million collection gap; 95 accounts below 80% collection	Approve a 90-day collections recovery plan with named account owners.
2. Protect high-value accounts	Top 20 accounts hold 24.7% of scored risk	Assign executive sponsors and account-specific retention actions.
3. Tighten discounts	$1.5 million discount leakage; 302 accounts exceeded 25%	Set approval thresholds and review the highest-impact representatives.
4. Rebuild leading indicators	Engagement fell from 66.7 to 52.0	Make engagement deterioration part of weekly account management.

The principal risk is not a lack of signal. The dataset already distinguishes payment, deterioration, churn, and discount risk at account level. The principal risk is execution: whether the business converts that signal into an owned queue, a short response cycle, and measurable recovery outcomes.



Context and objectives

Revenue leakage is the difference between the commercial value the business expects to earn and the cash it ultimately realizes. In this project, leakage emerges through discounting before billing and through payment failures after billing. Customer deterioration and silent churn are treated as forward-looking risk signals because they increase the probability that current ARR will not be retained.

The analysis answers four connected questions. First, how large is leakage and how has it changed? Second, which mechanisms explain the gap? Third, where is risk concentrated across customers, segments, regions, cohorts, and sales representatives? Fourth, which actions can reduce exposure within a realistic management cycle?

The output is designed for three audiences. Executives need a concise view of exposure and priorities. Commercial and finance leaders need an operating queue and control actions. Technical reviewers need transparent definitions, reproducible calculations, and explicit limitations.

Scope

The population comprises 500 B2B SaaS customers, 11,492 customer-month revenue records, and 60,000 customer-month-feature usage records. The observation window covers January 2024 through December 2025 without missing calendar months.

Data and methodology

Data architecture and unit of observation

Dataset	Rows	Unit of observation	Primary use
customers	500	Customer	Master attributes, ARR, churn status
monthly_revenue	11,492	Customer-month	Revenue funnel, payment and engagement trends
product_usage	60,000	Customer-month-feature	Usage context
risk_scorecard	500	Customer	Composite risk and prioritization
payment_analysis	500	Customer	Collection gaps and payment behavior
discount_analysis	500	Customer	Discount intensity and anomaly flags

The customer-month table is the core analytical fact table. Contracted MRR, billed MRR, collected MRR, payment status, engagement, and leakage pattern are observed together at that grain. Customer-level analytical tables aggregate those observations into risk indicators without changing the source tables.

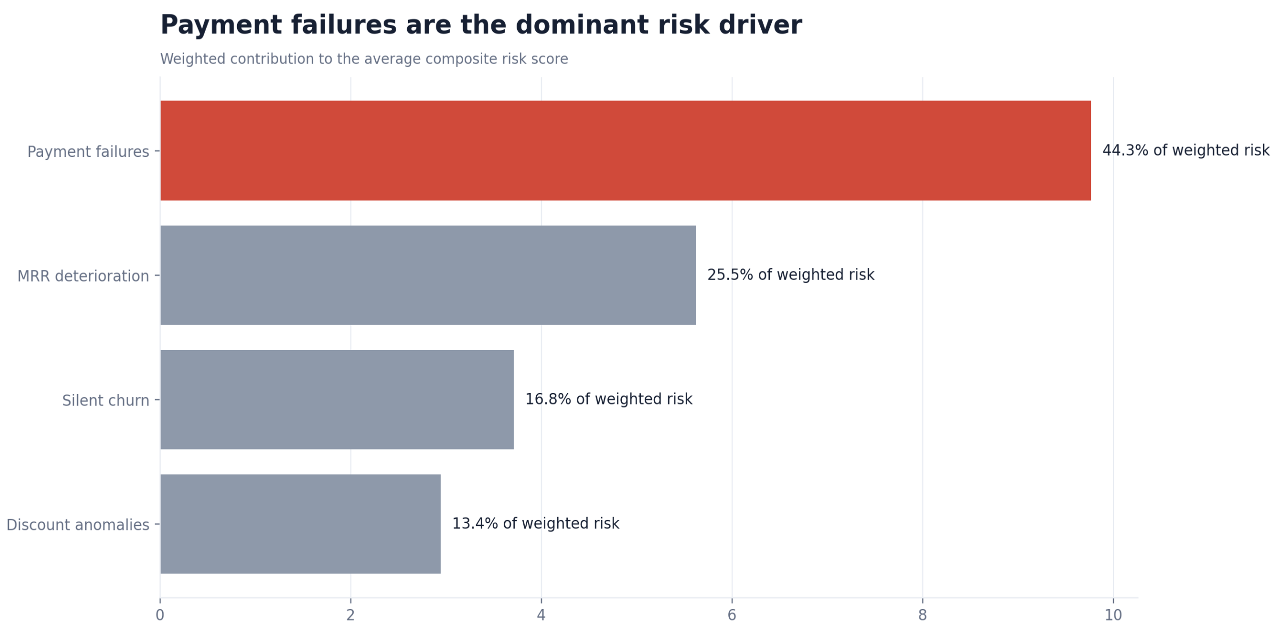
All calculations in this report use the processed project data. No external benchmarks or manually entered chart values are used.

Data and methodology

Revenue and risk definitions

Measure	Definition
Discount leakage	Contracted revenue minus billed revenue.
Collection leakage	Billed revenue minus collected revenue.
Total leakage	Contracted revenue minus collected revenue.
Collection rate	Collected revenue divided by billed revenue.
Composite risk score	30% churn + 25% deterioration + 20% discount + 25% payment risk.
Revenue at risk	Annual contract value multiplied by composite risk score.

The composite risk score is a prioritization device, not a probability of loss. Its weights are judgement-based and should not be interpreted as empirically calibrated causal effects. Revenue at risk therefore provides a consistent exposure ranking, but it is not a forecast of realized loss.



Data and methodology

Validation and quality controls

The project includes automated checks for row counts, required columns, null completeness on critical fields, score bounds, weighted-score reconciliation, revenue-at-risk caps, date coverage, and join safety. The processed tables reconcile to the source revenue funnel, and billed revenue does not exceed contracted revenue.

Two limitations in the data deserve explicit treatment. First, 73 scorecard engagement fields are structurally null for customers that exited before the recent lookback window. Second, one engagement observation exceeds the intended 0 to 100 scale. The single outlier has no material effect on aggregate findings but should be clamped in future generation runs.

Interpretation discipline

The analysis distinguishes realized leakage from prospective risk. Discounts and collection failures are realized gaps in the observed period. Churn, engagement deterioration, and composite revenue at risk indicate exposure that may or may not convert into loss. This distinction is maintained throughout the report to avoid overstating expected recovery.

Analytical framework

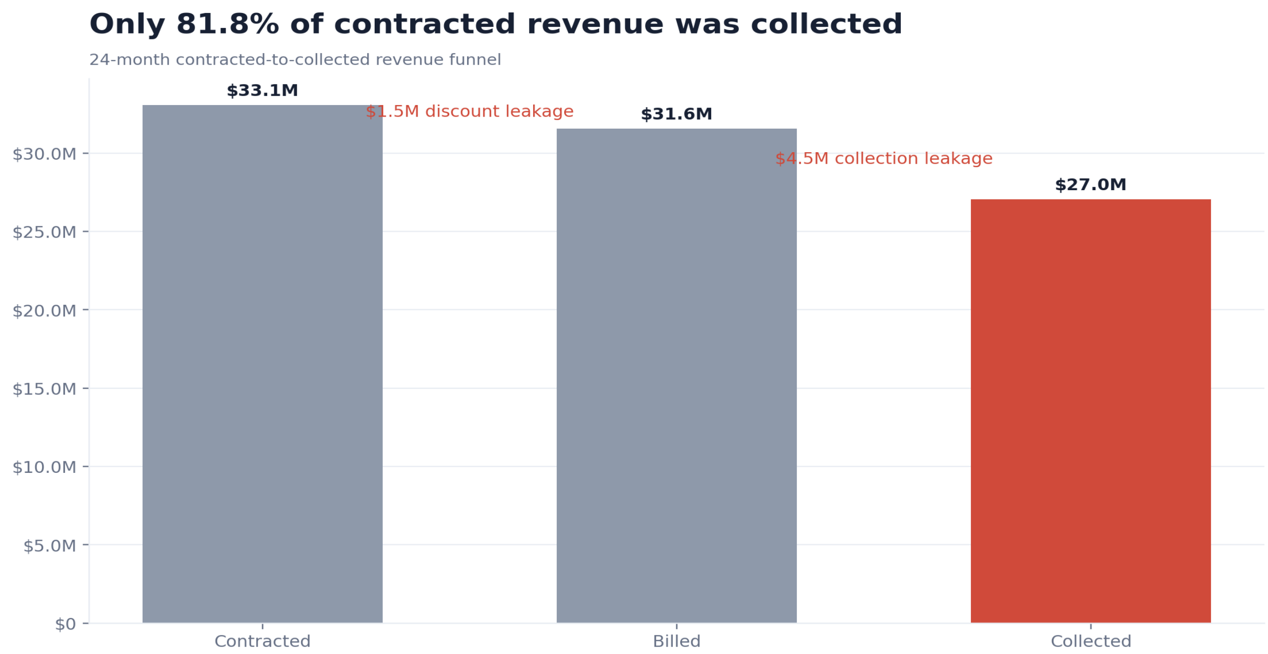
The analytical framework follows the revenue lifecycle. Contract value is established first. Discounting determines how much is billed. Payment behavior determines how much is collected. Usage and engagement provide earlier evidence of whether current revenue is durable. The scorecard combines those signals so management can rank accounts consistently.

Stage	Core question	Primary metric	Management owner
Contracting	Is price discipline holding?	Discount leakage	Sales leadership
Billing and collection	Is billed revenue realized?	Collection rate and gap	Finance and collections
Customer health	Is the account deteriorating?	Engagement and MRR trend	Customer success
Portfolio control	Where should intervention start?	Revenue at risk	Revenue leadership

This structure prevents a common analytical error: treating all leakage as a single undifferentiated number. The actions required to recover an overdue invoice differ from the actions required to reverse customer disengagement or govern discount approvals.

Finding 1: The contracted-to-collected gap is material

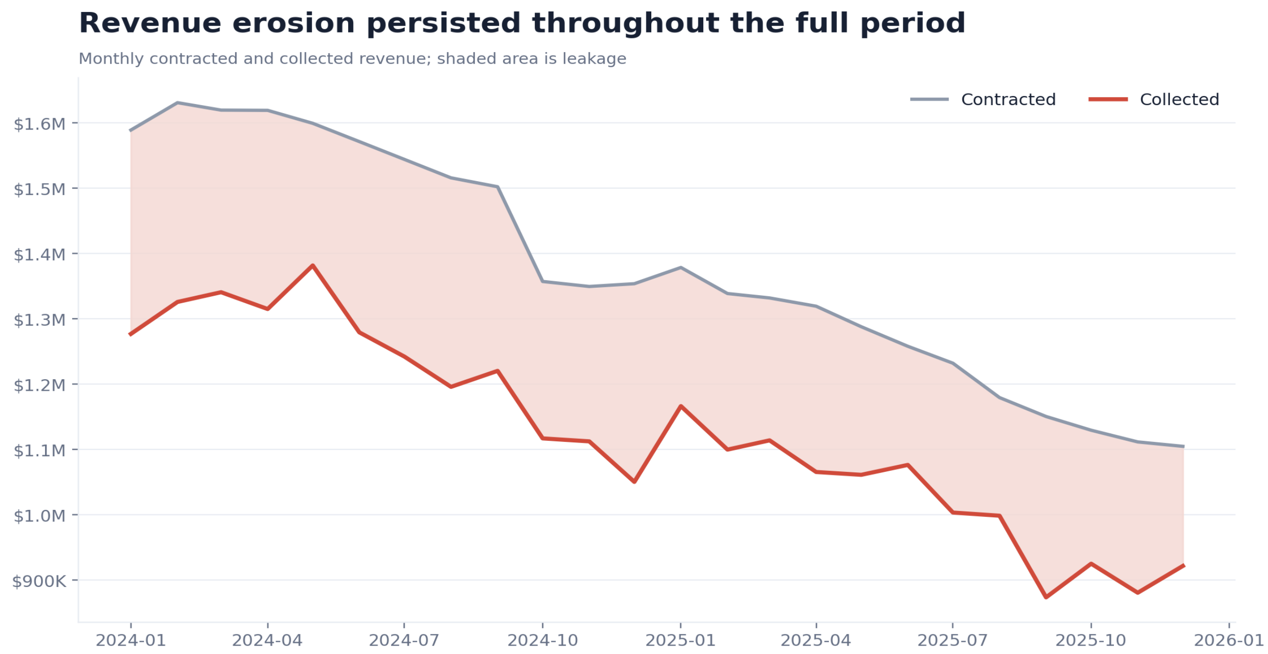
The business collected 81.8% of contracted revenue. The $6.0 million gap comprises $1.5 million lost before billing and $4.5 million lost after billing. Collection failures are 3.0 times the size of discount leakage, which makes cash realization the larger realized problem and the place where recovery effort earns the highest return per hour of management time.



What this changes: Set the recovery ambition against the $4.5 million collection gap first. A 20% reduction in that gap alone returns roughly $904,133 of cash, more than the entire discount problem is worth in a comparable period.

Finding 2: Leakage is persistent, not episodic

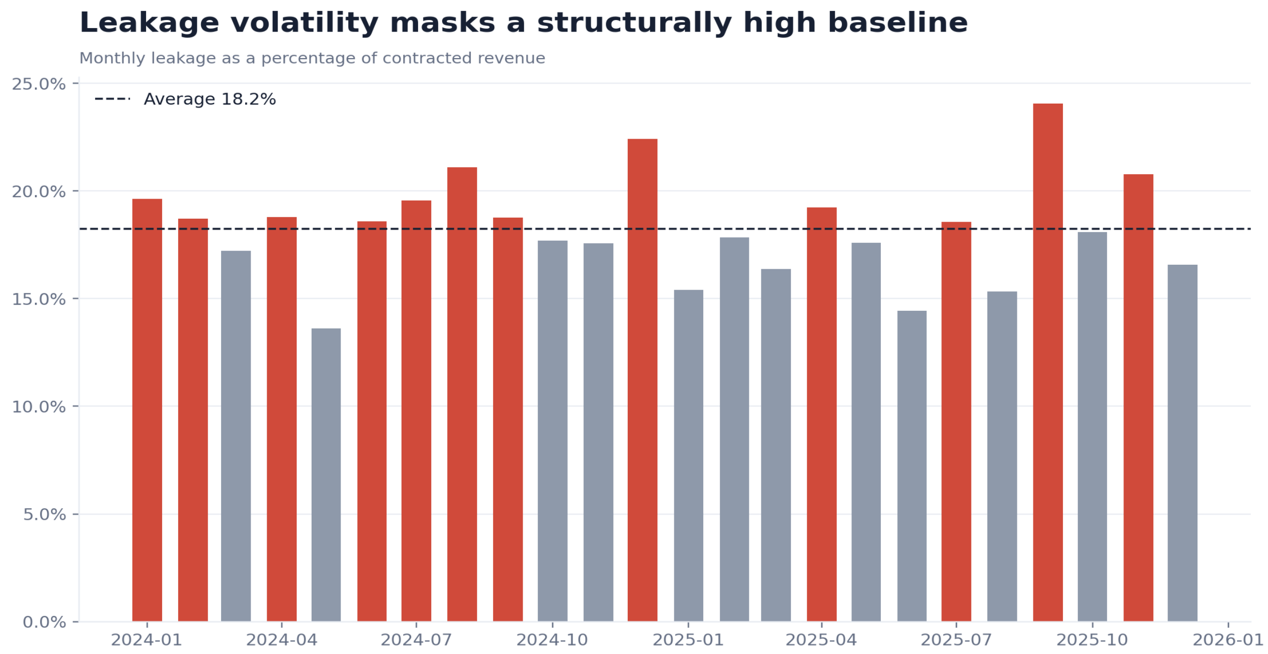
Contracted and collected revenue decline together over the period as the active customer base contracts, but the shaded gap between them is visible in every one of the 24 months. No single month carries the loss. The persistence rules out a one-quarter billing anomaly or a single large dispute as the explanation, and points instead to a standing weakness in how invoices convert to cash.



What this changes: Treat leakage as a recurring control to be governed monthly, not a backlog to be cleared once. A clean-up project will recover some cash and then the gap will reopen, because the mechanism that produces it every month has not changed.

Finding 3: Monthly volatility sits on a high baseline

The average monthly leakage rate is 18.2%. The last six months averaged 18.9% against 17.8% in the first six. Individual months swing above and below the mean, but every month clears a floor that is already too high. The swings invite month-by-month explanations that miss the structural point: even a good month leaks materially.



What this changes: Anchor reporting to the rolling baseline, not the latest month. A board that reacts to the 18.9% spike but relaxes at the next good month will never fund the standing fix the 18.2% average requires.

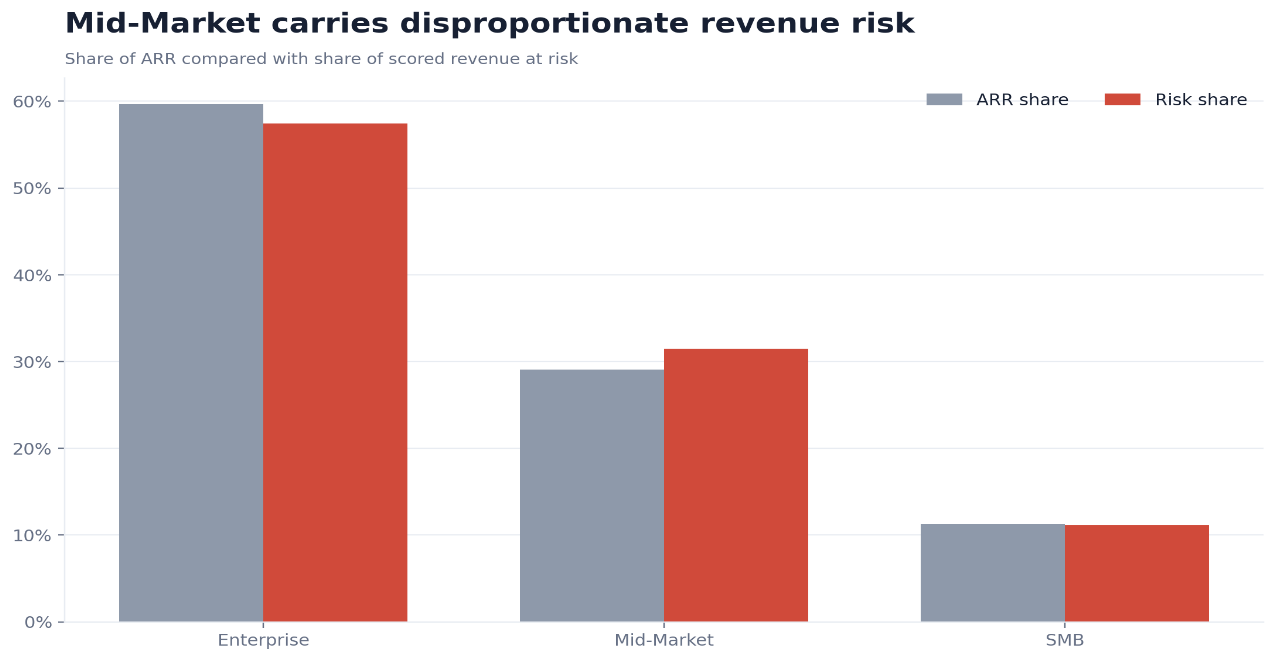
Finding 4: Payment risk dominates the composite score

Payment failures account for 44.3% of weighted composite risk, ahead of silent churn at 16.8% and MRR deterioration at 25.5%. Discount anomalies contribute 13.4%. The ranking is consistent with the realized funnel, where the billed-to-collected gap is 3.0 times the discount gap, so the model is pointing management at the same place the cash evidence already points.

What this changes: Weight the collections workstream above the others when allocating people. The risk model and the realized funnel agree, so there is no analytical reason to spread attention evenly across four drivers of unequal size.

Finding 5: Mid-Market risk is disproportionate

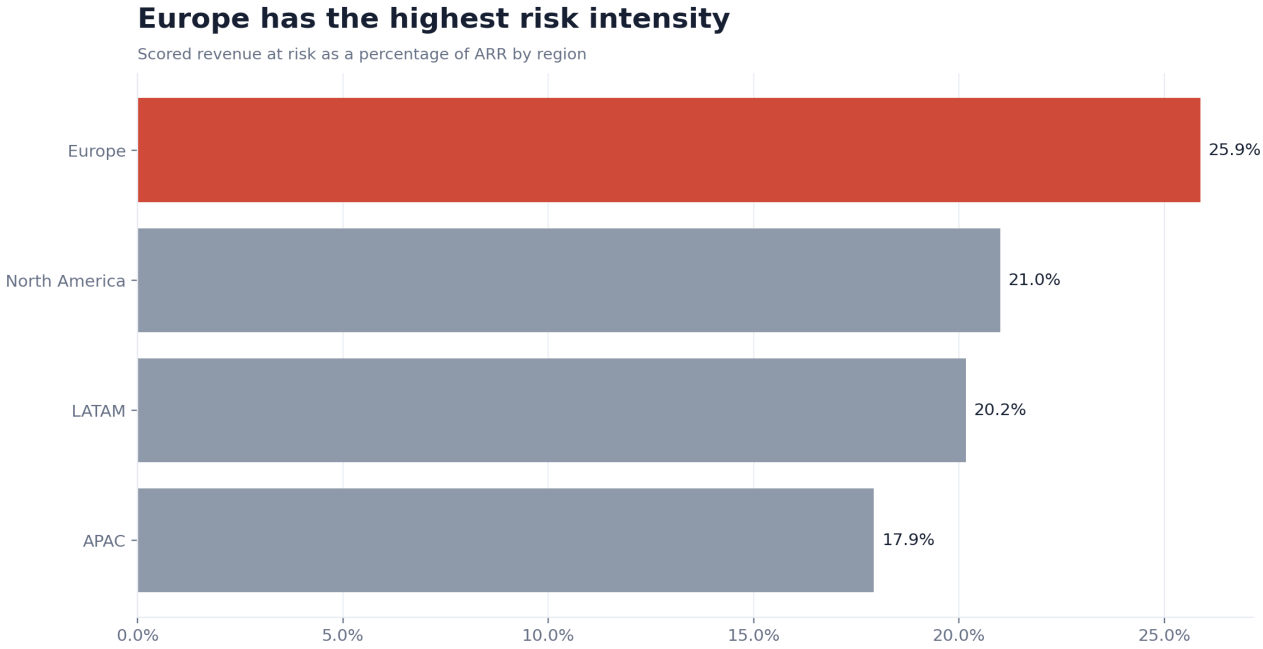
Enterprise holds the largest absolute exposure at $2.4 million because it carries most ARR. Mid-Market is the efficiency problem: it holds 31.4% of scored risk on 29.1% of ARR, a risk intensity of 1.08 times its fair share. Enterprise needs protection because of its size; Mid-Market needs a diagnosis because it is underperforming relative to the revenue it represents.



What this changes: Run a Mid-Market control review separate from the Enterprise account work. The two segments fail for different reasons, and a single playbook applied to both will over-serve large accounts and under-diagnose the segment that is actually inefficient.

Finding 6: Region risk intensity is uneven

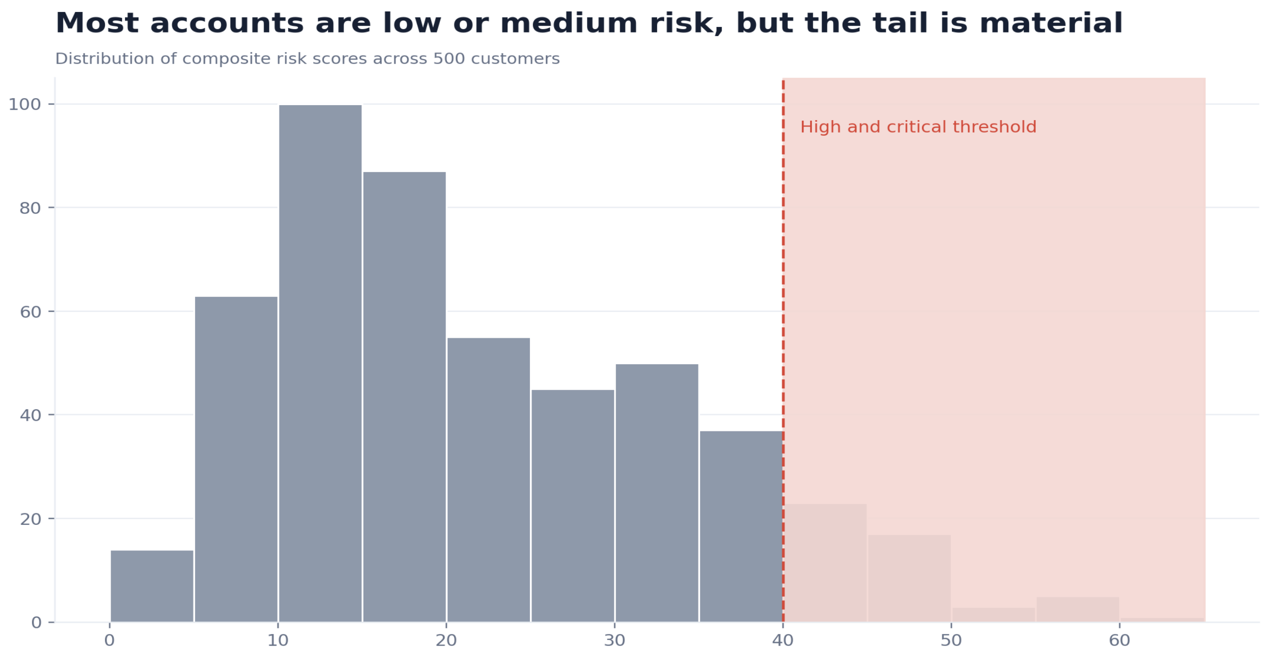
Europe has the highest scored revenue-at-risk rate at 25.9% of ARR, against 17.9% in APAC. The spread is wide enough that a single global collections standard would be too loose for the worst region and unnecessarily heavy for the best. Region is acting as a proxy for differences in account mix, payment terms, or local collections practice that the headline numbers hide.



What this changes: Commission a Europe collections diagnostic before rolling any standard out globally. The gap between Europe and APAC is too large to treat as noise, and the fix in one region may be the wrong fix in another.

Finding 7: The risk tail is operationally manageable

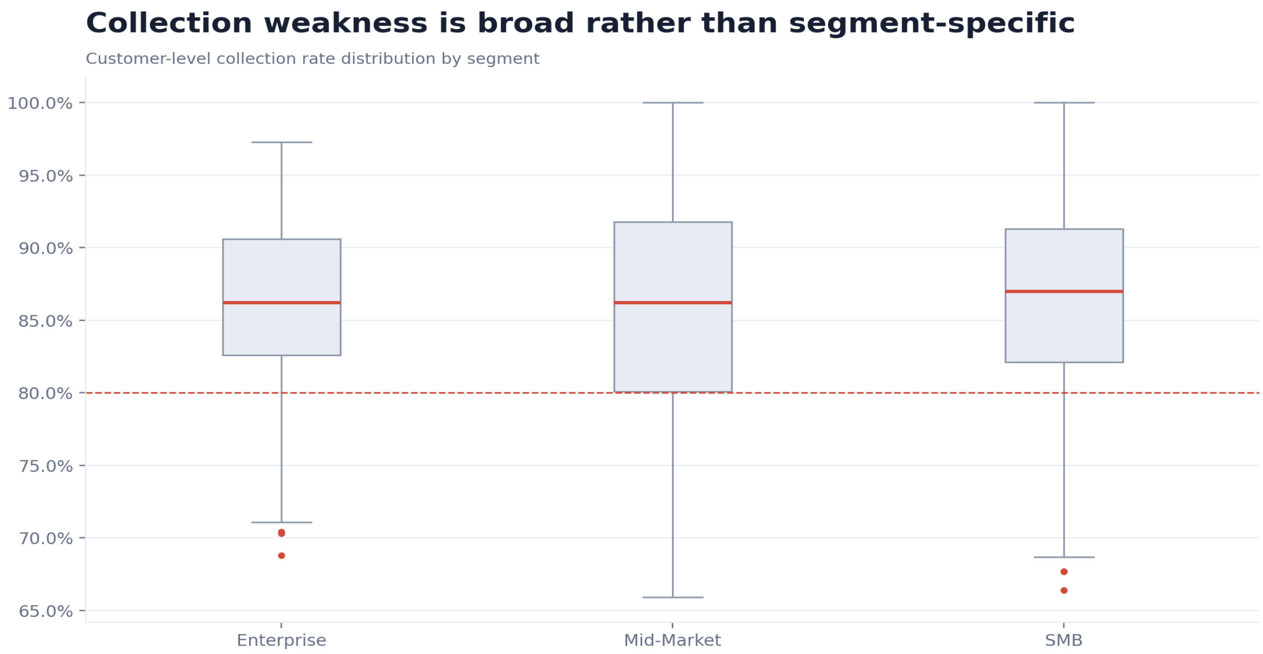
Only 48 of 500 accounts are critical or high risk, while 185 sit in the medium band. The distribution is reassuring and dangerous at once. The high-risk queue is small enough for named human ownership, but the medium band is large enough that a handful of accounts drifting upward each month would overwhelm that queue if no one is watching the migration.



What this changes: Assign the 48 high and critical accounts to named owners now, and put the 185 medium accounts under automated threshold monitoring. The medium band is a pipeline into the high-risk queue, not a resting state.

Finding 8: Collection weakness is broad

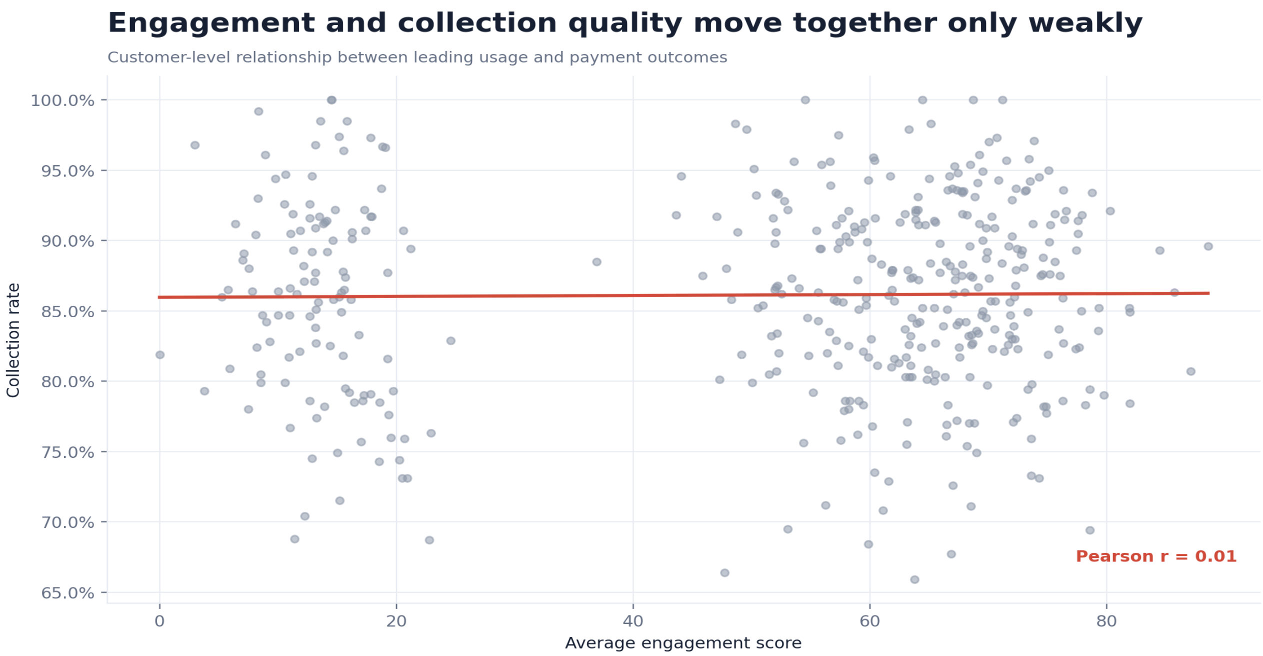
95 accounts collect less than 80% of what they are billed. The box plots overlap heavily across Enterprise, Mid-Market, and SMB, with weak collectors and the 80% line crossing every segment. There is no clean tier to quarantine. The problem is a property of the collections process, not of one customer type, which is why segment-only escalation rules would leave most of the weak accounts untouched.



What this changes: Build one collections operating model that applies to every segment, then layer segment-specific escalation on top. Designing the model around a single tier would miss the majority of the sub-80% accounts, which are spread across all three.

Finding 9: Engagement is useful but insufficient alone

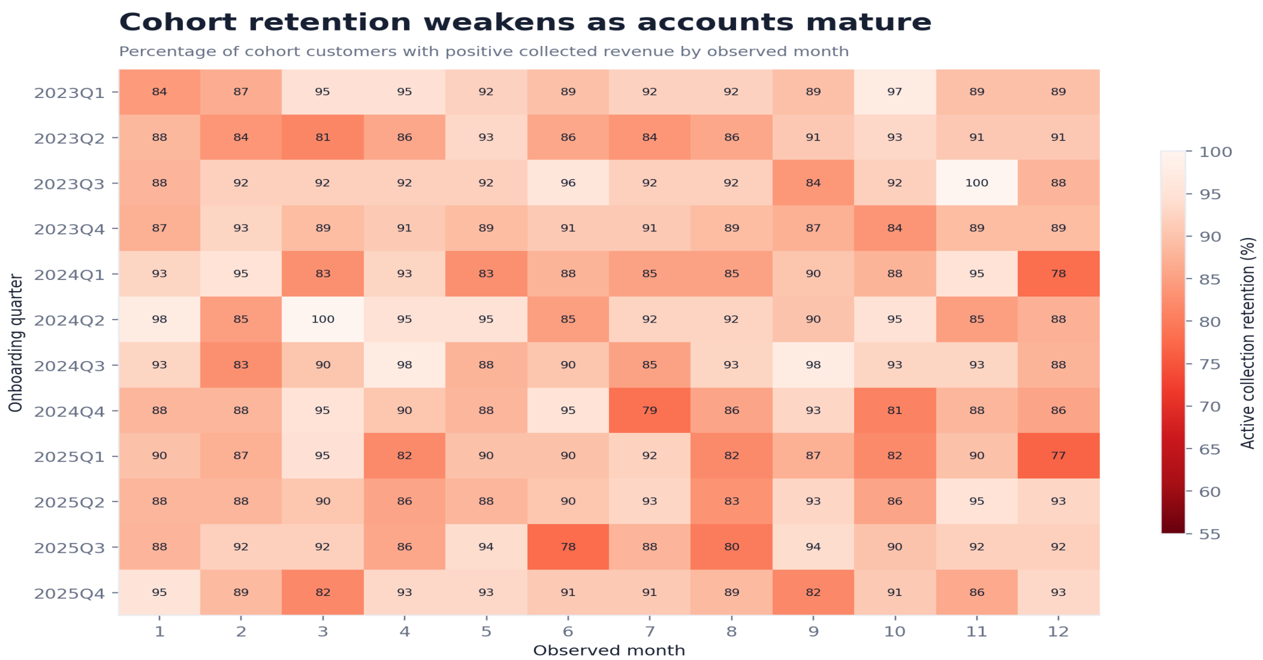
The customer-level link between engagement and collection quality is positive but weak. The scatter shows engaged accounts that pay poorly and disengaged accounts that pay on time. Engagement carries real early-warning signal, but it cannot stand in for payment evidence. A model that escalated on low engagement alone would chase healthy payers and miss quietly engaged accounts that have stopped paying.



What this changes: Keep engagement, payment status, and account economics as independent inputs to escalation. Collapsing them into one health score would discard the cases where the signals disagree, which are exactly the cases worth a human look.

Finding 10: Retention weakens with account maturity

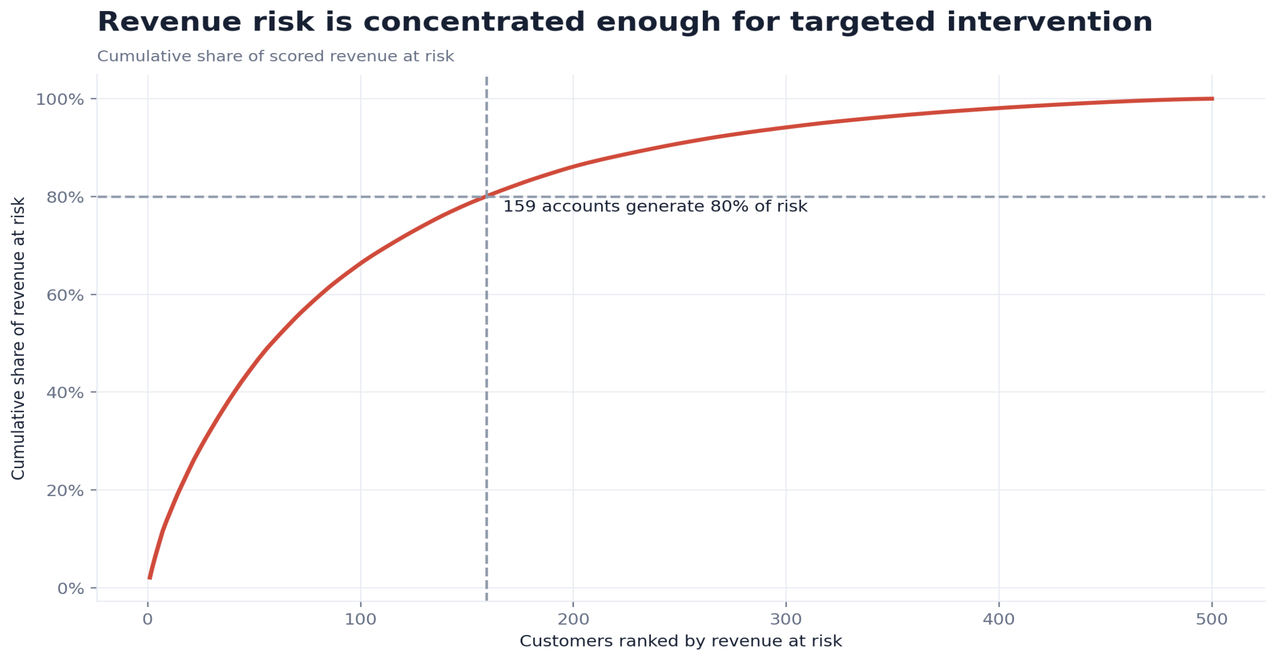
The cohort heatmap shows the share of each onboarding quarter still generating positive collected revenue falling as observed tenure grows. The decay is visible across cohorts, not confined to one weak intake. Because cohorts start at different calendar dates, the pattern is directional rather than a controlled experiment, but the consistency across cohorts makes a maturity effect the most credible reading.



What this changes: Trigger a retention checkpoint at the tenure month where the heatmap first turns, rather than waiting for an annual review. The decay is gradual and predictable, which means it can be intercepted on a schedule instead of discovered at renewal.

Finding 11: Revenue risk is concentrated

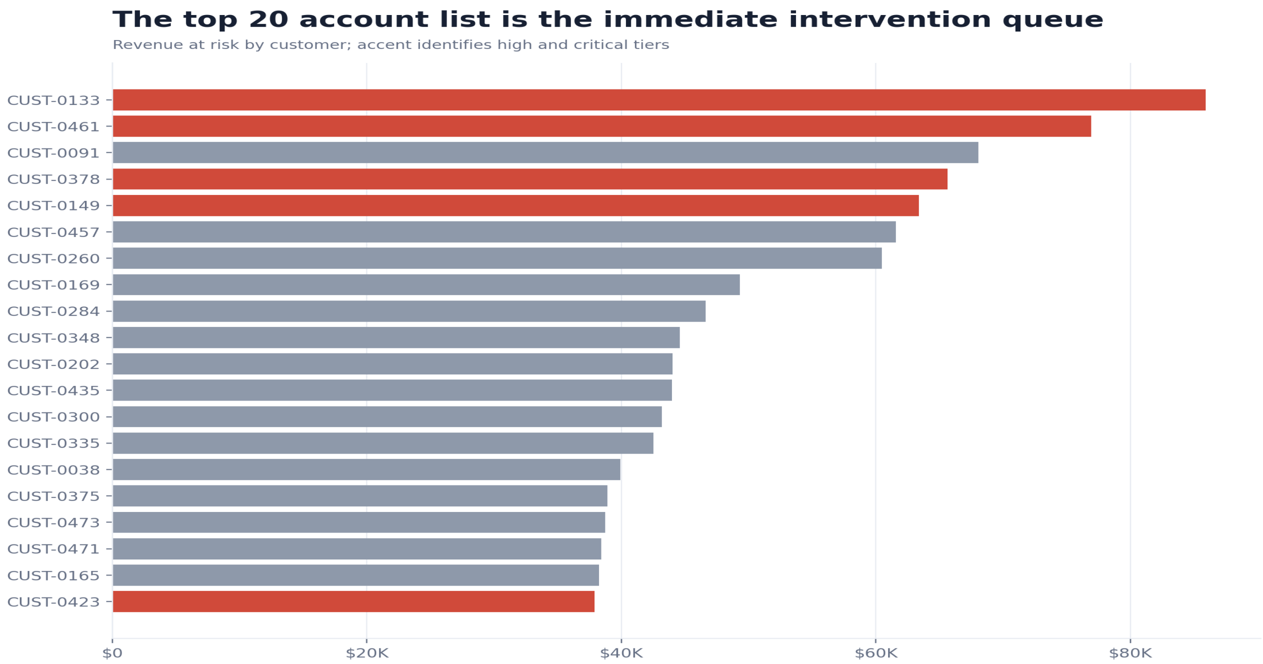
The top 20 accounts hold 24.7% of scored revenue at risk and the top 50 hold 45.6%. The Pareto curve climbs steeply and then flattens, which is the shape that makes targeted intervention worthwhile. A small, nameable set of accounts carries most of the exposure, so management capacity spent at the top of the list is not diluted across the long tail.



What this changes: Cap the immediate intervention list at the accounts that make up the first 46% of cumulative risk and resource it fully, rather than spreading thin attention across all 500. The curve says the marginal account beyond the top 50 returns very little.

Finding 12: The immediate queue is identifiable

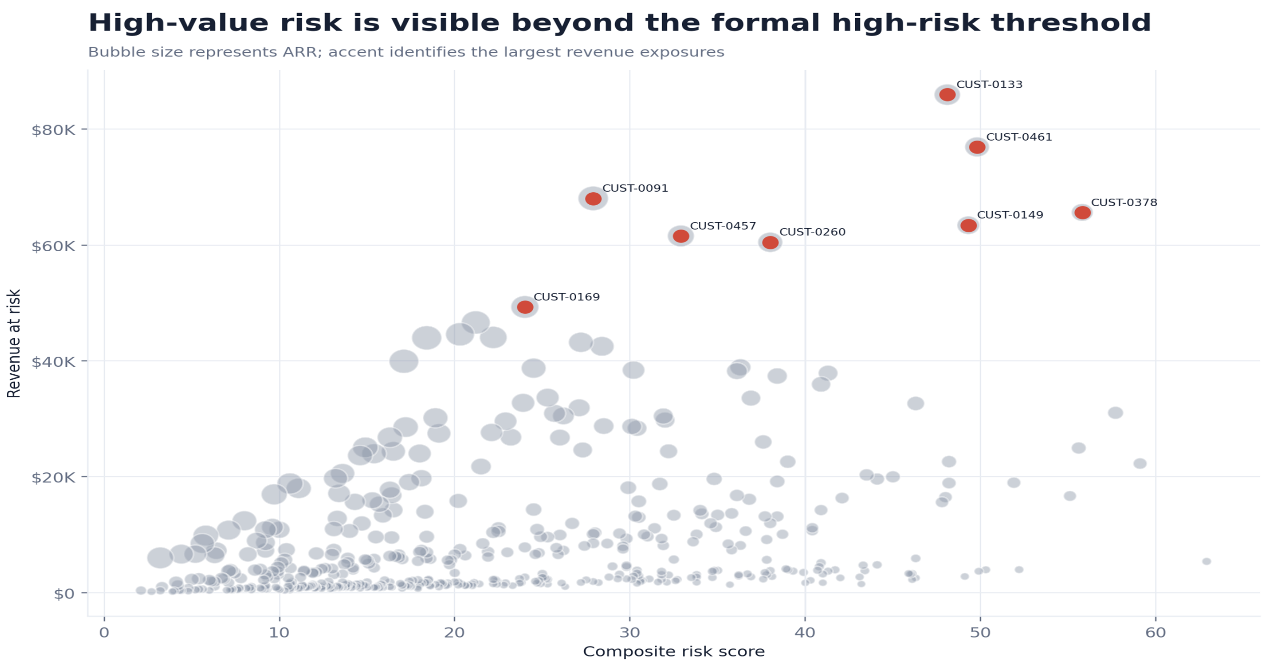
The largest exposures cluster among accounts that combine high contract value with medium-to-high composite scores, and the accent bars mark the high and critical tiers within that list. Ranking by revenue at risk rather than by score alone is what surfaces this queue: a mid-score account on a large contract can outrank a high-score account on a small one, and it is the dollar exposure that should set the order of work.



What this changes: Work the top-20 list in revenue-at-risk order and assign an executive sponsor to each. Sorting the same accounts by risk score would reorder the queue and push some of the largest dollar exposures down the page.

Finding 13: Formal tier thresholds do not capture all value exposure

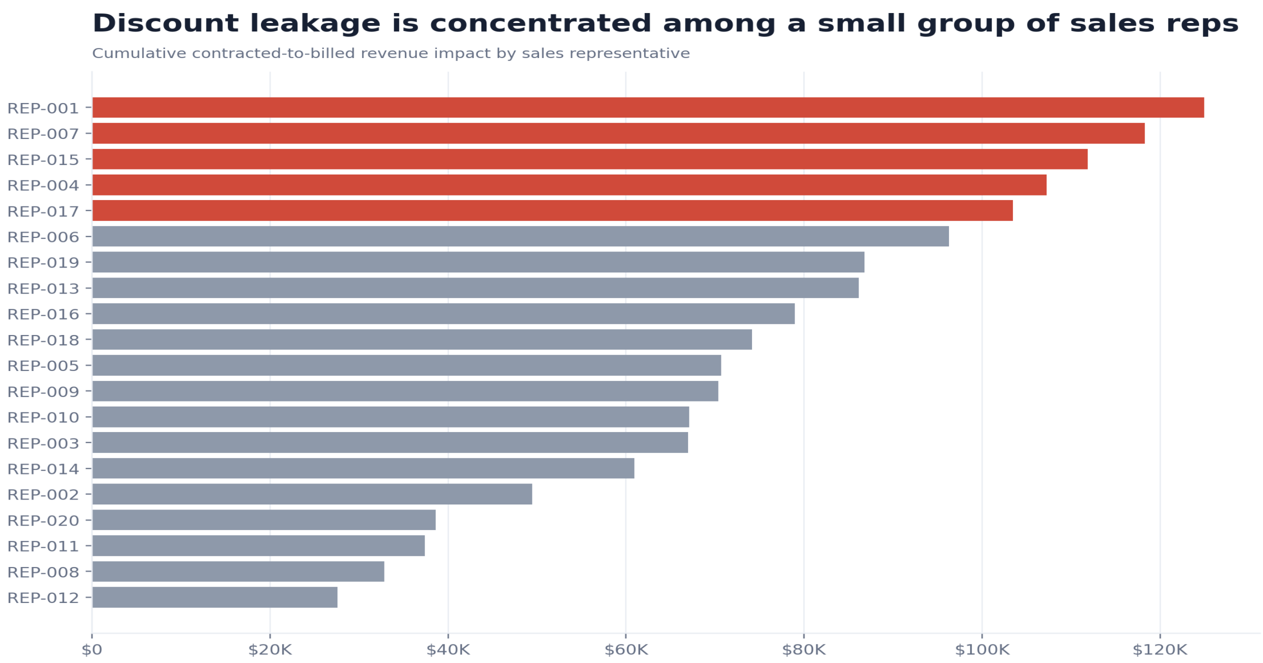
On the prioritization matrix, several large bubbles sit to the left of the high-risk line yet still carry substantial revenue at risk. A pure tier cut-off would route these high-value, medium-risk accounts to passive monitoring. The matrix shows why a single threshold is the wrong control: value and risk are separate axes, and an account can be dangerous on one while looking safe on the other.



What this changes: Run two parallel queues, one keyed on risk intensity and one on absolute value exposure, and review any account that appears on either. A single threshold optimised for risk will systematically under-serve the high-value medium-risk corner.

Finding 14: Discount leakage has clear ownership

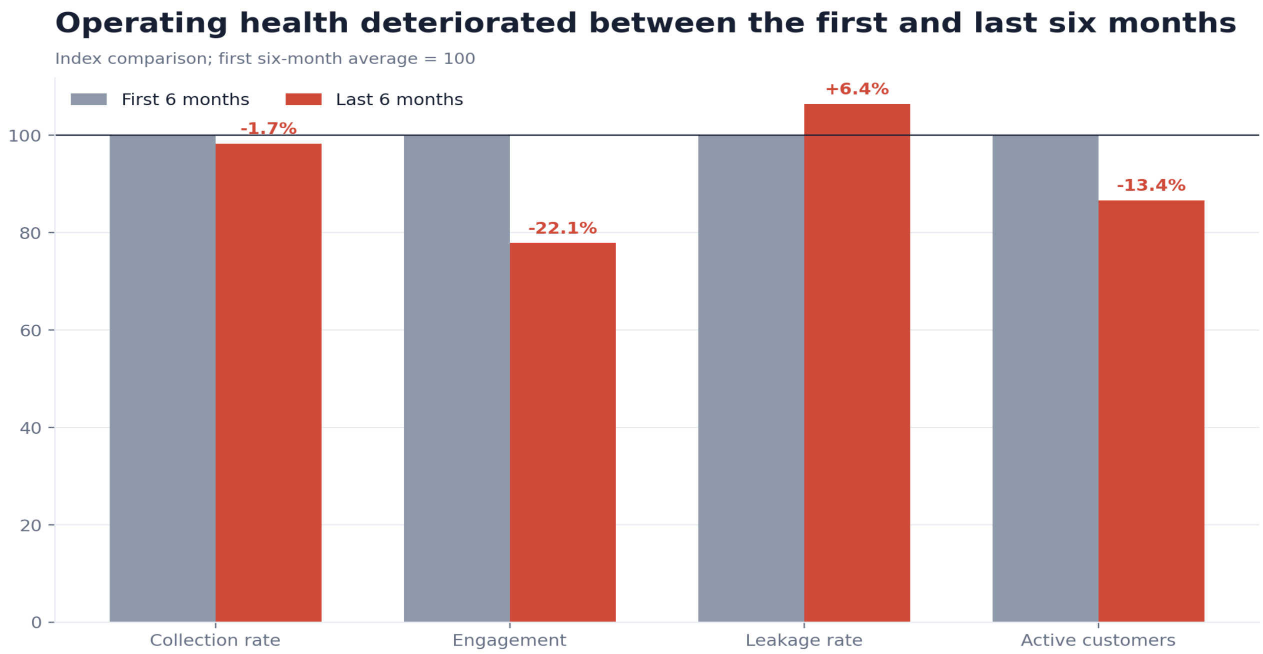
The highest-impact representative, REP-001, accounts for $124,986 of contracted-to-billed leakage, and the accent bars mark the top quartile that drives most of the total. Across the book, 302 accounts carry a maximum discount above 25%. Unlike collection failures, this leakage is created at a known desk by a known decision, which makes it the most directly governable line in the funnel.



What this changes: Require named approval and a recorded rationale for any discount above 25%, and review the top-quartile representatives monthly. Because the loss is concentrated and authored, a policy change at a handful of desks closes most of the $1.5 million before it reaches billing.

Finding 15: Operating health deteriorated

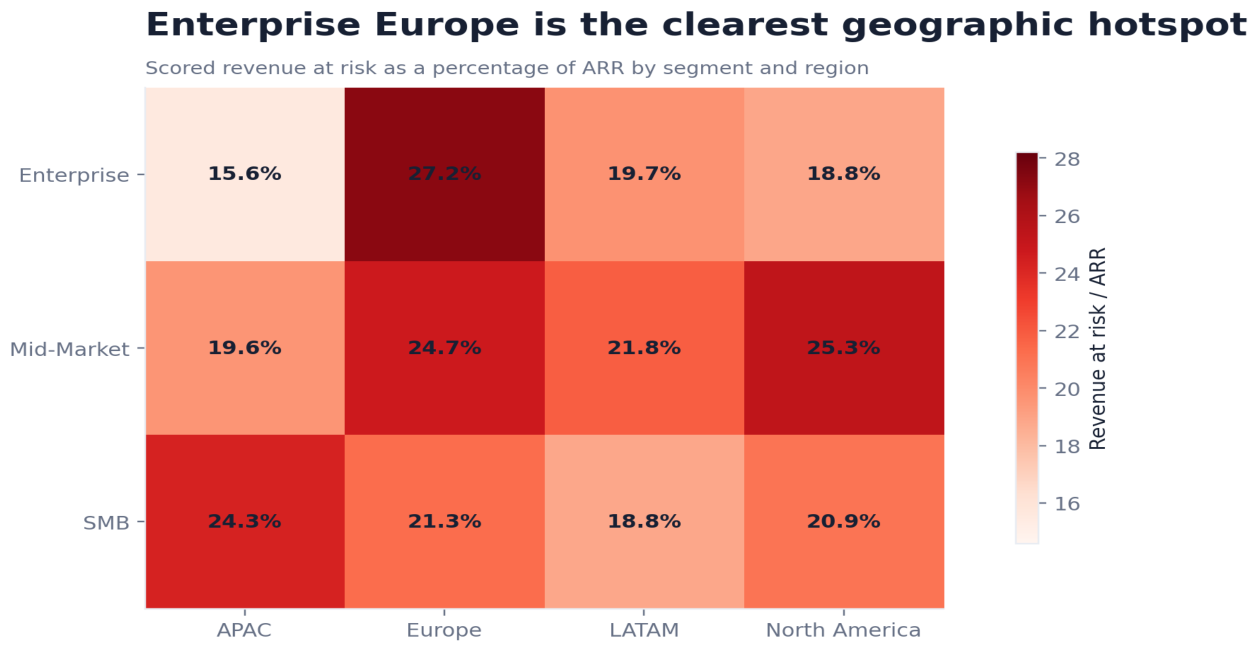
Between the first and last six months, average collection rate fell from 86.3% to 84.8%, average engagement from 66.7 to 52.0, and the active customer count from 500 to 433. Three independent measures moved the same way at the same time. Any one could be explained away; together they describe a base that is shrinking and paying less reliably as it shrinks.



What this changes: Stand up a monthly operating-health review that escalates when collection, engagement, and customer count deteriorate together. Simultaneous movement across independent measures is a stronger signal than any single metric and should carry more weight in the room.

Finding 16: Enterprise Europe is the clearest hotspot

The segment-region heatmap puts Enterprise Europe at the top of the intensity scale, combining the high contract values of the Enterprise segment with the elevated risk rate of the region. This is the one cell where scale and intensity coincide, which is what separates a genuine hotspot from a large segment that is merely large or a risky region that is merely small.



What this changes: Create a single Enterprise Europe recovery workstream with one consolidated view of its largest accounts. Managing it through the generic regional and segment forums would split accountability for the one exposure that most warrants a dedicated owner.

Risks, limitations, and caveats

Model and data limitations

Limitation	Implication	Required safeguard
Modelled data	Patterns demonstrate methodology but do not represent a live business.	Do not use the amounts as external benchmarks or operating forecasts.
Judgement-based score weights	Risk scores are rankings, not calibrated probabilities.	Back-test against realized churn and cash recovery before deployment.
Point-in-time scorecard	Trajectory is simplified into current scores.	Add recurring score snapshots and migration analysis.
No causal identification	Associations do not prove intervention effects.	Measure outcomes against explicit control or comparison groups.
Limited external context	Macroeconomic and market signals are absent.	Integrate credit, market, and support data where lawful and useful.

The largest interpretive risk is to treat revenue at risk as expected loss. It is a weighted exposure measure. The report uses it to prioritize attention, while realized leakage remains anchored to the contracted, billed, and collected revenue fields.

Risks, limitations, and caveats

Operational risks

A recovery programme can create unintended customer harm if collection escalation ignores account context or if discount controls slow legitimate strategic deals. Controls should therefore distinguish avoidable exceptions from approved commercial policy. Every override should have an owner, reason, value, and expiry date.

A second risk is metric gaming. Sales teams may avoid discounts by changing contract structures; collections teams may improve short-term rates by delaying recognition of difficult accounts. Management should review the complete contracted-to-collected funnel and monitor changes in definitions, not only headline KPIs.

A third risk is queue overload. The high-risk queue is manageable, but the medium-risk population is large. Automated monitoring should handle medium-risk accounts, with human intervention triggered by worsening score migration, material value, or repeated payment failure.

Recommendations and action priorities

First 30 days: establish control and ownership

Action	Owner	Measure of completion
Create a top-20 recovery queue with executive sponsors.	Revenue leadership	All top-20 accounts have an owner, action, value target, and next review date.
Launch a collections sprint for accounts below 80% collection.	Finance	Every account has a root-cause code and recovery disposition.
Introduce approval for discounts above 25%.	Sales leadership	100% of new exceptions include approver, reason, and expiry.
Publish a weekly contracted-to-collected control pack.	BI and finance	Metrics reconcile to source data and show movement by owner.

The first month should focus on operating discipline rather than model sophistication. The business already has enough signal to identify material accounts and prevent additional uncontrolled discounting.

Recommendations and action priorities

Days 31 to 90: convert the analysis into a management system

Action	Design requirement	Target outcome
Account risk reviews	Review high-risk accounts weekly and high-value medium-risk accounts monthly.	Reduce unowned high-value exposure to zero.
Collections segmentation	Separate disputes, process failures, credit risk, and customer distress.	Improve collection rate and shorten time to resolution.
Discount governance	Track requested, approved, and realized discount by representative and deal type.	Reduce preventable discount leakage without slowing strategic deals.
Score migration	Store monthly score snapshots and monitor tier movement.	Detect deterioration before accounts enter the high-risk queue.

The operating system should measure both avoided leakage and customer outcomes. Recovery that damages retention may improve one metric while worsening total economics.

Recommendations and action priorities

Measurement framework

KPI	Definition	Cadence	Decision use
Total leakage rate	(Contracted - collected) / contracted	Monthly	Overall control effectiveness
Collection recovery	Cash recovered from targeted overdue accounts	Weekly	Collections sprint progress
Discount exception value	Contract value lost through approved exceptions	Weekly	Pricing governance
High-value exposure	Revenue at risk in top-value account queue	Weekly	Executive intervention
Risk migration	Accounts moving up or down risk tiers	Monthly	Leading indicator quality
Intervention yield	Recovered or retained value / targeted exposure	Monthly	Programme effectiveness

Targets should be set after one clean baseline cycle. Premature targets based on unreconciled or incomplete history would create false precision. The first objective is consistent measurement; the second is sustained improvement.

Recommendations and action priorities

Decision rights and governance

Revenue leadership should own the total leakage outcome. Finance should own collection controls and realized cash recovery. Sales leadership should own discount governance. Customer success should own engagement deterioration and retention actions. BI should own metric definitions, reconciliation, and the reliability of the intervention queue.

Forum	Frequency	Required decisions
Recovery stand-up	Weekly	Account actions, escalations, and expected recovery dates
Commercial control review	Monthly	Discount exceptions, segment and regional hotspots, score migration
Executive review	Quarterly	Leakage trend, realized recovery, retention impact, and policy changes

The governance model should remain small. Additional meetings add value only when they change account ownership, policy, or resource allocation.

Appendix A: headline metrics

Metric	Value
Annual recurring revenue	$19.2 million
24-month contracted revenue	$33.1 million
24-month billed revenue	$31.6 million
24-month collected revenue	$27.0 million
Total leakage	$6.0 million (18.2%)
Scored revenue at risk	$4.2 million (21.7% of ARR)
Collection rate	85.7%
High and critical accounts	48
Top 20 share of risk	24.7%

Appendix B: segment summary

Segment	Customers	ARR	Revenue at risk	Risk intensity
Enterprise	89	$11.5 million	$2.4 million	0.96x
Mid-Market	152	$5.6 million	$1.3 million	1.08x
SMB	259	$2.2 million	$464,090	0.99x

Risk intensity compares each segment's share of scored revenue at risk with its share of ARR. Values above 1.00 indicate disproportionate exposure.

Appendix C: regional summary

Region	Customers	ARR	Revenue at risk	Risk rate
APAC	103	$3.9 million	$692,620	17.9%
Europe	148	$5.6 million	$1.4 million	25.9%
LATAM	53	$1.8 million	$364,322	20.2%
North America	196	$8.0 million	$1.7 million	21.0%

Appendix D: payment status economics

Payment status	Records	Billed	Collected	Gap	Collection rate
pending	606	$1.9 million	$0	$1.9 million	0.0%
failed	579	$1.5 million	$0	$1.5 million	0.0%
partial	903	$2.6 million	$1.5 million	$1.1 million	57.3%
paid	8,022	$21.8 million	$21.8 million	$0	100.0%
paid_late	1,382	$3.7 million	$3.7 million	$0	100.0%

Appendix E: top 25 accounts by revenue at risk

Customer	Segment	Region	Risk tier	Score	Revenue at risk
CUST-0133	Enterprise	North America	High	48.1	$85,939
CUST-0461	Enterprise	Europe	High	49.8	$76,921
CUST-0091	Enterprise	North America	Medium	27.9	$68,036
CUST-0378	Enterprise	Europe	High	55.8	$65,620
CUST-0149	Enterprise	North America	High	49.3	$63,404
CUST-0457	Enterprise	Europe	Medium	32.9	$61,557
CUST-0260	Enterprise	Europe	Medium	38.0	$60,461
CUST-0169	Enterprise	APAC	Medium	24.0	$49,305
CUST-0284	Enterprise	North America	Medium	21.2	$46,619
CUST-0348	Enterprise	North America	Medium	20.3	$44,585
CUST-0202	Enterprise	Europe	Medium	22.2	$44,039
CUST-0435	Enterprise	North America	Low	18.4	$43,974
CUST-0300	Enterprise	North America	Medium	27.2	$43,200
CUST-0335	Enterprise	LATAM	Medium	28.4	$42,510
CUST-0038	Enterprise	Europe	Low	17.1	$39,922
CUST-0375	Enterprise	Europe	Medium	36.3	$38,894
CUST-0473	Enterprise	Europe	Medium	24.5	$38,742
CUST-0471	Enterprise	LATAM	Medium	30.2	$38,405
CUST-0165	Enterprise	Europe	Medium	36.1	$38,235
CUST-0423	Enterprise	Europe	High	41.3	$37,878
CUST-0286	Enterprise	Europe	Medium	38.4	$37,386
CUST-0292	Enterprise	APAC	High	40.9	$35,953
CUST-0459	Enterprise	North America	Medium	25.3	$33,659
CUST-0355	Enterprise	LATAM	Medium	36.9	$33,569
CUST-0043	Enterprise	Europe	Medium	23.9	$32,754

Appendix: calculation notes

All monetary totals are calculated directly from the project CSV files. Revenue funnel totals sum monthly contracted, billed, and collected revenue. Segment and regional exposure summaries aggregate the customer-level risk scorecard. Charts use the same calculated fields as the report narrative.

The report intentionally excludes decorative visuals and redundant charts. Each visual answers a distinct analytical question and is also delivered as an individual PNG in the graph pack.

Methodology notes

The chart pack, dashboard, and report are generated from the validated datasets in the repository. The dashboard embeds its charting library and does not require a network connection. Analysis can be reproduced by running the pipeline from source.